Signals & Systems Solutions — Solutions (SIGNALS-SYSTEMS-HS2018)

Q#	Question / Context	Selected Answer	Solution Logic
q1	Stability condition for forward Euler discretization of dy/dt = a y + u	**The discrete-time system is stable iff a < 0 and 0 < Ts < -2/a.**	*Forward Euler yields y[k+1] = (1 + a Ts) y[k] + Ts u[k]; pole p = 1 + a Ts. Stability requires |p| < 1 => -1 < 1 + a Ts < 1 => -2 < a Ts < 0. For a > 0 no positive Ts satisfies; for a = 0 pole =1 (marginally stable); for a < 0 gives 0 < Ts < -2/a.*
q2	When Euler and exact discretizations of dy/dt = a y + u coincide	**Euler and exact discretization correspond only when a = 0 (or Ts = 0).**	*Exact ZOH discretization: y[k+1] = e^{aTs} y[k] + ((e^{aTs}-1)/a) u[k] (a≠0). Euler gives y[k+1] = (1 + aTs) y[k] + Ts u[k]. Equality of coefficients requires e^{aTs}=1 + aTs and Ts = (e^{aTs}-1)/a, which holds only if aTs = 0. With Ts>0, this implies a=0. For a=0 both reduce to y[k+1]=y[k]+Ts u[k].*
q1	Analyze linearity, stability, causality, time-invariance, and memorylessness of the system y[n] = a y[n-3] + u[n] + 0.5 cos(2π n) as a varies.	**Not linear for any a; BIBO stable iff |a| < 1; causal for all a; time-invariant for all a; memoryless only when a = 0.**	*The constant 0.5 term breaks homogeneity, so the system is nonlinear for all a. The homogeneous part y_h[n] = a y_h[n-3] has poles at a^{1/3} e^{j2πk/3}; stability requires |a|^{1/3}<1 → |a|<1. The equation uses only y[n-3] and u[n], so it is causal and time■invariant for any a. Memorylessness requires output to depend only on the current input; this holds only when a=0, giving y[n]=u[n]+0.5.*
q2	Find the output of G2 in the cascade u[n] → G1 (h1[n]=sin(8n)) → G2 (h2[n]=a^n s[n], |a|<1) when the input to the cascade is u[n]=δ[n]–aδ[n–1].	**y[n] = sin(8n).**	*The overall impulse response is h = h1 * h2. Using convolution sums, h[n] = (1/D)[(1–a cos8) sin(8n) – a sin8 cos(8n)] with D=1–2a cos8+a^2. Convolving with u[n] gives y[n]=h[n]–a h[n–1]. Substituting h[n] and simplifying using trig identities yields y[n]=sin(8n), independent of a (for |a|<1).*
q5	Find the output of the cascade of two LTI systems with impulse responses h1[n]=sin(8n) and h2[n]=a^n u[n] (|a|<1) when the input is u[n]=δ[n]-aδ[n-1].	**y[n] = sin(8n)**	*The overall impulse response is h[n]=h1[n]*h2[n]. Using DTFT, H(e^{jΩ}) = H1(e^{jΩ})H2(e^{jΩ}) where H2(e^{jΩ})=1/(1-a e^{-jΩ}) and H1(e^{jΩ}) corresponds to sin(8n). This yields h[n] = (sin(8n)-a sin(8n+8))/(1-2a cos8 + a^2). The output y[n]=u[n]*h[n]=h[n]-a h[n-1] simplifies to sin(8n) after algebraic manipulation using trigonometric identities.*
q6	Given x[n] nonzero only for positive even n as 1/n^2, define y[n]=x[n-1] via frequency shift, q[n]=x[n]+x[n+1], and r[n]=q[n]-q[n-2] for even n>0 (zero otherwise). Determine r[n].	**r[n] = x[n] - x[n-2] for n even >0, zero otherwise; equivalently r[2]=1/4, r[n]=1/n^2 - 1/(n-2)^2 for n even ≥4, and r[n]=0 otherwise.**	*Multiplication by e^{jΩ} shifts the signal: y[n]=x[n+1]. Hence q[n]=x[n]+x[n+1]. For even n>0, x[n+1]=0 and x[n-1]=0, so r[n]=q[n]-q[n-2]=x[n]-x[n-2]. Since x[n]=1/n^2 for positive even n and zero otherwise, we obtain r[2]=1/4-0=1/4 and for n≥4 even, r[n]=1/n^2 - 1/(n-2)^2. For odd n or n≤0, r[n]=0.*
q7	Determine if z[n] = sum_{k=0}^{N-1} x[k]/y[n-k] is periodic given x[n], y[n] periodic with period N and y[n] != 0.	**Yes, z[n] is periodic with period N.**	*Define w[n] = 1/y[n]; then z[n] = sum_{k=0}^{N-1} x[k] w[n-k] is the circular convolution of x[n] and w[n]. Since both x[n] and w[n] are periodic with period N, their circular convolution is also periodic with period N. Hence z[n] repeats every N samples.*
q8	Express the DFS coefficients Z[k] of z[n] in terms of X[k] and W[k], where W[k] are DFS coefficients of w[n]=1/y[n].	**Z[k] = X[k] * W[k].**	*The DFS of a circular convolution of two length-N sequences equals the product of their DFS coefficients. Since z[n] is the circular convolution of x[n] and w[n], we have Z[k] = X[k] W[k] for k = 0,...,N-1.*
q9	Design a causal, stable LTI filter G that outputs the DC component c0 for input u[n]=c0 + c1(-1)^n + c2 cos(pi/3 n).	**G(z) = (1/2)(1 + z^{-3}) (equivalently y[n] = 0.5[u[n] + u[n-3]])**	*The filter must have unity gain at ω=0 and zero gain at ω=π and ω=±π/3. Placing zeros at z=-1, e^{±jπ/3} gives numerator (z+1)(z^2 - z +1)=z^3+1. Scaling for unit DC gain yields H(z)=(1/2)(1+z^{-3}). This FIR filter is causal and stable.*
q10	Find a1 for the causal stable system H(z)=b0/(1+a1 z^{-1}) when input u[n]=cos(π/2 n) produces output with DFS coefficients Y[1]=3.2+j1.6, Y[3]=3.2-j1.6.	**0.5**	*Input DFS coefficients are U[1]=U[3]=0.5. Hence H(e^{jπ/2})=Y[1]/U[1]=2*(3.2+j1.6)=6.4+j3.2. With H(e^{jω})=b0/(1+a1 e^{-jω}), at ω=π/2 we get b0/(1-ja1)=6.4+j3.2. Solving gives a1=0.5 and b0=8.*
q11	Find b0 for causal stable LTI system H(z)=b0/(1+a1 z^{-1}) given input u[n]=cos(πn/2) and output DFS coefficients Y[k]={0,3.2+j1.6,0,3.2-j1.6}.	**8**	*Input is cos(πn/2) = (e^{jπn/2}+e^{-jπn/2})/2. Its DFS coefficients are U[1]=U[3]=0.5. Output Y[1]=3.2+j1.6 gives H(e^{jπ/2})=Y[1]/U[1]=6.4+j3.2. With H(e^{jω})=b0/(1+a1 e^{-jω}) and e^{-jπ/2}=-j, solving b0/(1-ja1)=6.4+j3.2 yields a1=0.5 and b0=8.*
q12	Determine if y[n]=2e^{jn} can be output of an LTI system with input u[n]=e^{jn}, and if such system is unique.	**Yes, such a system exists (e.g., y[n]=2x[n]), but it is not unique because only the frequency response at ω=1 is constrained.**	*For an LTI system, output to complex exponential e^{jωn} is H(e^{jω}) e^{jωn}. Here H(e^{j1}) must equal 2. Many systems (e.g., pure gain 2, or any filter with H(e^{j1})=2) satisfy this, so a system exists but is not uniquely determined by this single input-output pair.*
q13	Check whether y■[n]=2cos(n)s[n] can be the LTI output for input u■[n]=cos(n)s[n]; if possible, determine uniqueness.	**Yes, e.g., y[n]=2x[n]; not unique.**	*The input is cos(n) for n≥0 and zero otherwise; the output is exactly twice the input for n≥0 and zero otherwise. A memoryless gain of 2 (h[n]=2δ[n]) is LTI and produces this output. However, the input contains only frequencies ω=±1 rad/sample. Any LTI system whose frequency response satisfies H(e^{j±1})=2 will yield the same output, so the system is not uniquely determined by this single input■output pair.*

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q14	Given that u■[n] and y■[n] are both periodic (see figure), decide if y■[n] could be the LTI output of u■[n]; comment on uniqueness.	**Yes, e.g., y[n]=u[n]; not unique in general.**	*For an LTI system, a periodic input produces a periodic output with the same period. If the figure shows u■[n] and y■[n] sharing the same period, an LTI system can exist (e.g., the identity system). Uniqueness would require the input to excite all frequency components (i.e., contain a full set of harmonics) so that H(e^{jω}) is determined at those frequencies. Since only one periodic pair is given, the system is not uniquely defined.*
q15	Determine if y[n] can be output of an LTI system given input u[n]; both periodic (figure). If possible, comment on uniqueness.	**Such a system exists iff for every harmonic where the input's Fourier coefficient is zero the output's coefficient is also zero; the ratio Y_k/U_k must be identical for all harmonics with nonzero input. If these conditions hold, a system exists; it is unique only if the input has nonzero coefficients at all harmonics (i.e., excites every frequency line), otherwise the system is not unique because H can be arbitrary at frequencies where input is zero.**	*For an LTI system, $Y(e^{j\omega}) = H(e^{j\omega})U(e^{j\omega})$. With periodic signals, only discrete frequencies $\omega_k = k \cdot 2\pi/N$ appear. Hence for each k, $Y_k = H_k U_k$. If $U_k = 0$ then Y_k must be 0; otherwise $H_k = Y_k/U_k$ must be same for all k with $U_k \neq 0$. If any $U_k=0$ and $Y_k \neq 0$, no LTI system can produce that output. When condition satisfied, H_k is forced at frequencies where $U_k \neq 0$; at frequencies where $U_k=0$, H_k can be chosen arbitrarily, leading to non-uniqueness unless there are no such frequencies.*